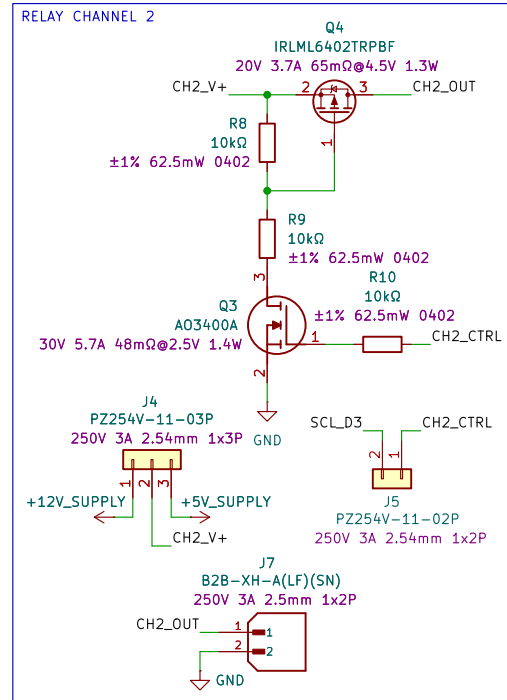
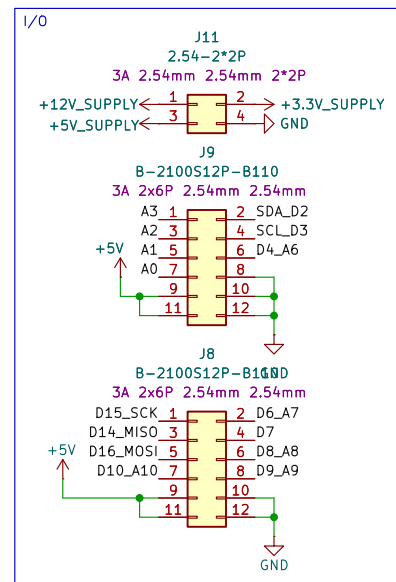
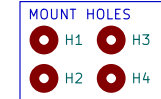
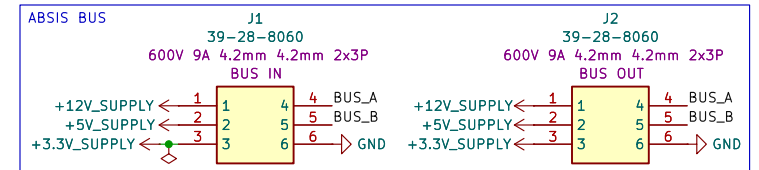


- 1) RS485 BUS TERMINATION: ON THE LAST SLAVE DEVICE OF AN RS485 BUS DO NOT INSTALL J2 (BUS OUT) MOLEX CONNECTOR. INSTEAD, INSTALL A 120 Ω THRU-HOLE RESISTOR ACROSS PINS 4 AND 5 OF J2 (BUS OUT).
- 2) ABIS RS485 BUS SPECIFICATION
 - 2.1) BUS IS HALF DUPLEX; TRANSMIT OR RECEIVE ONLY AT ANY GIVEN TIME.
 - 2.2) PIN FUNCTION: R = DATA OUTPUT / D = DATA INPUT / DE HAS INTERNAL PULL-DOWN (DRIVER DISABLED BY DEFAULT) / RE HAS INTERNAL PULL-UP (RECEIVER DISABLED BY DEFAULT).
 - 2.3) CIRCUIT PROVIDES OPEN, SHORT, AND IDLE-BUS FAILSAFE OPERATION TO PREVENT ELECTRICAL NOISE FROM BEING INTERPRETED AS VALID DATA.
 - 2.4) THVD1410DR TRANSCIVER OPERATES WITH TTL-LEVEL INPUTS. LOGIC HIGH: 2 V TO VCC / LOGIC LOW: 0 V TO 0.8 V.
 - 2.5) CAUTION: MAX DATA RATE DECREASES AS CABLE LENGTH INCREASES.

- 3.1) ALE+ IS DESIGNED FOR NOM 12 V ABSIS BUS PWR. DO NOT APPLY 12 V TO XU2 VCC OR ANY LOGIC/I/O SIGNAL.
- 3.2) IC2 REGULATES THE 12 V BUS TO APPROX 7.3 V. XU2 RAW IS SUPPLIED FROM THE +7.3 V RAIL THRU D2.
- 3.3) D2 IS A SERIES SILICON RECTIFIER DIODE BETWEEN THE +7.3 V RAIL AND XU2 RAW. DO NOT BYPASS OR SHORT D2.
- 3.4) D2 PROVIDES REVERSE-CURRENT ISOLATION BETWEEN XU2 RAW AND THE +7.3 V RAIL. D2 IS NOT A 12 V CLAMP OR OVERVOLTAGE PROTECTION DEVICE.
- 3.5) DO NOT BACKFEED THE +7.3 V RAIL OR XU2 VCC FROM EXT EQUIPMENT. USB PWR, WHEN USED, SHALL ENTER THRU THE PROMICRO USB CONNECTOR.
- 3.6) F1, A RESETTABLE FUSE, AND D3, A 12 V TVS DIODE, FORM THE 12 V INPUT PROTECTION CIRCUIT. DO NOT BYPASS OR SUBSTITUTE THESE DEVICES WITH JUMPERS DURING NORMAL OPERATION.
- 3.7) XU2 VCC IS THE PROMICRO +5 V LOGIC SUPPLY RAIL. DO NOT USE XU2 VCC AS A SUBSTITUTE FOR THE +7.3 V OR +12 V PWR RAILS.

- 4.1) J8 AND J9 ARE UNKEYED 2X6, 2.54 MM GENERAL-PURPOSE I/O HEADERS. VERIFY PIN 1 ORIENTATION BEFORE CONNECTING HARNESS OR TEST EQUIPMENT.
- 4.2) J8 AND J9 I/O SIGNALS CONNECT DIRECTLY TO PROMICRO I/O PINS. EXT SIGNAL VOLTAGE SHALL NOT EXCEED THE LOGIC SUPPLY VOLTAGE OF THE INSTALLED PROMICRO. I/O SIGNALS ARE NOT 12 V TOLERANT.
- 4.3) DO NOT DRIVE J8 OR J9 I/O SIGNALS WHILE U1 IS UNPOWERED. EXT SIGNALS MAY BACKPOWER U1 THRU THE MICROCONTROLLER INPUT PROTECTION CIRCUITRY.
- 4.4) ALL EXT DEVICES CONNECTED TO J8 OR J9 SHALL SHARE COMMON GND WITH ALE+ UNLESS GALVANIC ISOLATION IS PROVIDED EXTERNALLY.
- 4.5) SIGNAL NAMES SDA_D2, SCL_D3, D4_A6, AND D6_A7 IDENTIFY MULTIPLEXED PROMICRO FUNCTIONS. VERIFY FIRMWARE PIN ASSIGNMENT BEFORE CONNECTING EXT HARDWARE.
- 4.6) VCC AT THE I/O HEADERS IS A LOGIC SUPPLY AND SHALL NOT BE USED TO PWR HIGH-CURRENT LOADS.
- 4.7) J8 AND J9 ARE INTENDED FOR SHORT, INTERNAL I/O CONNECTIONS. PROVIDE APPROPRIATE LEVEL SHIFTING, ESD PROTECTION, TRANSIENT PROTECTION, OR ISOLATION FOR EXT OR LONG-CABLE APPLICATIONS.



Title: ABSIS ALE (ALmost Everything) PLUS

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